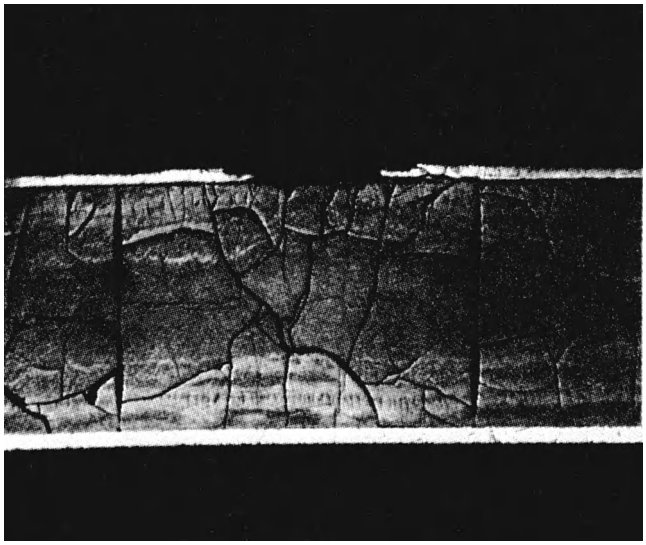


Figure 8.4 Zircaloy canned UO_2 fuel after exposure to very high power (high dose and heat) in a test reactor. The pellets have fractured and the canning has been penetrated. (SKI report)

research) yields ozone, while radiation of humid air yields HNO_3. One of the first observations of radiation-induced changes was the darkening of glass. Glass often contains iron, manganese, and other metals that can exist in several oxidation states with different colours. As a result of the irradiation, the oxidation state can change, resulting in change in colour. Dislocations as well as trapped electrons also contribute to the colour changes in glass. In chemical and metallurgical work with highly active substances it is desirable to observe the experiment through a thick glass window, which provides protection from the radiation. In order to avoid the colouring of the glass, a small amount (1−2%) of an element which can act as an electron trap is added, e.g. CeO_2, which acts by the reaction $Ce^{4+} + e^- \rightarrow Ce^{3+}$. After an exposure of 10^4 Gy the transmission to light of ordinary glass had been reduced to 44%, while for a CeO_2 protected glass it was still 89%.

Glass is very resistant to radiation damage because it is a non-crystalline "solid" liquid. Therefore, in such a material, it is not possible to speak of dislocations: the random structure of the glass allows it to include foreign species throughout the sample. This is why glass has been intensely studied, including full scale tests, as the matrix ("container") for high active waste (HAW) consisting of fission products and actinides.

8.6. WATER

The consequences of ionization and excitation depend on the physical state and the molecular composition of the irradiated material. In this section we introduce chemical phenomena into the description using the radiolysis of water as an example.

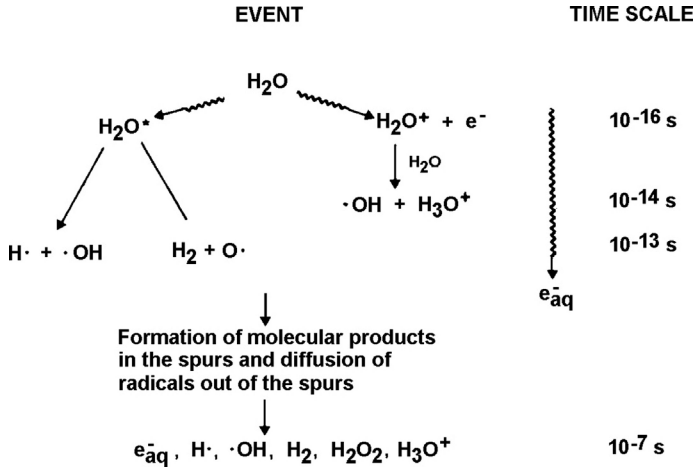


Figure 8.5 Time scale of radiolysis of water.

The time scale for the sequence of events on the radiolysis of water is shown in Figure 8.5. The ionization event occurs on the time scale of an electronic transition ($<10^{-16}$ s). The positive ion H_2O^+ formed reacts with water within 10^{-14} s, forming an $\cdot OH$ radical and H_3O^+. The electron, if liberated with sufficient kinetic energy, can ionize further water molecules before its energy falls below the ionization threshold of water (12.61 eV). The electron thereafter loses the rest of its energy by causing vibrational and rotational excitation of the water molecules and, finally, becomes solvated. The solvated electron is one or the most powerful reducing agents.

The solvation process has been shown to occur within 10^{-12} s. The excited states dissociate within 10^{-14}–10^{-13} s, i.e. on the same time scale as a molecular vibration to form $O\cdot$, $H\cdot$, $\cdot OH$ and H_2. The physical and physicochemical (pre-thermal) processes are thus completed within 10^{-12} s leaving the species in thermal equilibrium with the water.

The radiolysis products are clustered in spurs (Fig. 8.1); i.e., they are inhomogeneously distributed in the water and proceed to diffuse out of the spur volume. During this "spur diffusion" process, recombination reactions take place leading to the formation of molecular or secondary radical products. The spur expansion is complete within 10^{-7} s, at which time the radiolysis products are those shown in Figure 8.5. Spur reactions are listed in Table 8.2. *G*-values for the radiolysis products in water irradiated with different types of radiation are given in Table 8.3. It is seen that, as the *LET* of the radiation increases, the *G*-values of radical products decrease while the molecular *G*-values increase, which evidences the formation of tracks. The higher *G*-values for water consumption and for molecular product formation in irradiated water vapour, as compared to liquid water, show that spur formation and spur reactions are of minor importance in the

Table 8.2 Spur reactions in water

$e_{aq}^- + e_{aq}^- \rightarrow H_2 + 2OH^-$
$e_{aq}^- + \cdot OH \rightarrow OH^-$
$e_{aq}^- + H_3O^+ \rightarrow H\cdot + H_2O$
$e_{aq}^- + H\cdot \rightarrow H_2 + OH^-$
$H\cdot + H\cdot \rightarrow H_2$
$\cdot OH + \cdot OH \rightarrow H_2O_2$
$\cdot OH + H\cdot \rightarrow H_2O$
$H_3O^+ + OH^- \rightarrow H_2O$

Table 8.3 Product yields (μmol/J) in irradiated neutral water

Radiation	G(-H_2O)	G(H_2)	G(H_2O_2)	G(e_{aq}^-)	G(H·)	G(·OH)	G(·HO_2)
Liquid water:							
γ and fast electrons	0.43	0.047	0.073	0.28	0.062	0.28	0.0027
12 MeV He^{2+}	0.294	0.115	0.112	0.0044	0.028	0.056	0.007
Water vapor:							
γ,electrons	0.85	0.05	0	$G(e^-) = 0.31$	0.75	0.85	

vapour phase. This is to be expected in a less dense medium where the average distance between separate spurs will be larger. Also the diffusivities will be larger in the vapour phase.

In the absence of solutes, reactions between radical and molecular species occur in the bulk water. In pure water irradiated with γ or X-rays, water is reformed via the reactions

$$H_2 + \cdot OH \rightarrow H\cdot + H_2O \tag{8.16}$$

$$H\cdot + H_2O_2 \rightarrow H_2O + \cdot OH \tag{8.17}$$

$$\cdot OH + H\cdot \rightarrow H_2O \tag{8.18}$$

and no net decomposition of water is observed. The yields depend on the *LET*-value of the radiation, as illustrated in Figure 8.6 for acidic and neutral water. Irradiation with higher *LET* radiation, which have higher *G*-values for molecular products than for radical products, causes net decomposition of water. The reason for this is that with a high LET more radiolysis products are formed close to each other and may thus react forming longer lived molecular species. This will also have the effect that e.g. alpha radiation is more prone to cause oxidation but gamma radiation is more prone to cause a reductive environment.

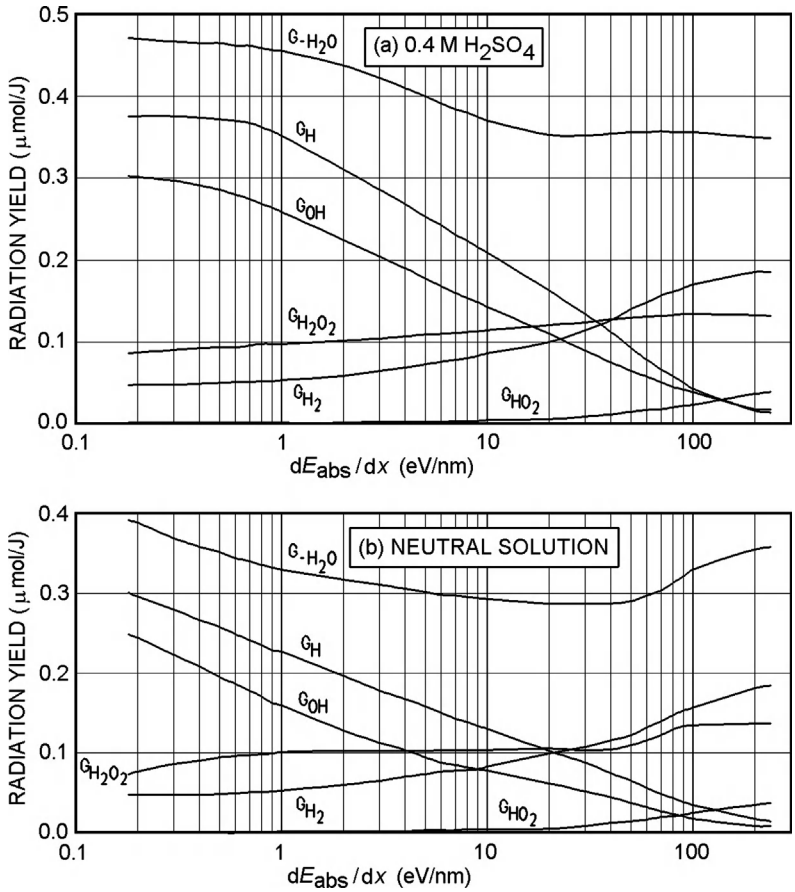


Figure 8.6 *G*-values (μmol/J) for radiolysis of water as a function of the *LET* value (eV/nm) of the system: (a) 0.4 M H_2SO_4, and (b) neutral solution. *(From A. O. Allen.)*

In nuclear reactors water used as a coolant or moderator should be as pure as possible to minimize dissociation during the time in the reactor. The formation and accumulation of an explosive gas mixture of H_2 and O_2 must be carefully avoided in all reactors in order to prevent accidents. Moreover the decomposition products of water can increase the corrosion of fuel elements, structural material, etc. Many reactors use N_2 as a protective gas. In this case the radiolysis can lead to the formation of HNO_3 unless suppressed by an excess of H_2 which preferentially yields NH_3. The pH of the water may be regulated by the H_2(g) pressure. The problem here is obvious though. Reactor water chemistry is monitored by taking samples and then measuring composition etc when cooler. By then the most reactive radiolysis products have already reacted and thus no real information of the corrosivity close to, or in, the reactor core can be obtained.

8.7. AQUEOUS SOLUTIONS

In irradiated dilute aqueous solutions at concentrations <0.1 mol/l, practically all the energy absorbed is deposited in the water molecules. Hence, the observed chemical changes are the result of the reactions between the solutes and the products of the water radiolysis. With increasing solute concentrations, the direct radiolysis of the solute gradually becomes important and the solute may also interfere with the spur reactions. The use of high concentrations of solutes which react selectively with the radical products — so-called *scavengers* — have provided experimental support for the existence of spurs.

In dilute solutions the chemical changes of a specific solute primarily reflect its reactivity towards e_{aq}^-, $H\cdot$ and $\cdot OH$. The hydrated electron e_{aq}^- is a strongly reducing species ($E_0 = -2.9$ V) whereas the hydrogen atom is a less powerful reductant ($E_0 = -2.3$ V). The H atom can be considered as a weak acid with a pKa of 9.6

$$e_{aq}^- + H^+ \rightleftharpoons H\cdot \tag{8.19}$$

The hydroxyl radical $\cdot OH$ is a strong oxidant ($E_0 = 2.7$ V in acidic and 1.8 V in basic solution). In strongly alkaline solution $\cdot OH$ dissociates into its anionic form O^- which has nucleophilic properties

$$\cdot OH \rightleftharpoons \cdot O^- + H^+ (pKa = 11.9) \tag{8.20}$$

In oxygenated solutions the perhydroxyl radical $\cdot HO_2$ is formed in reactions between e_{aq}^-, $\cdot H$ and O_2

$$e_{aq}^- + O_2 \rightarrow \cdot O_2^- \tag{8.21}$$

$$H\cdot + O_2 \rightarrow \cdot HO_2 \tag{8.22}$$

$$\cdot HO_2 \rightleftharpoons \cdot O_2^- + H^+ \ (pKa = 4.7) \tag{8.23}$$

In saline waters (0.1 M Cl^-, pH 4–10) the dominant radiolysis reactions are

$$\cdot OH + Cl^- \rightleftharpoons HClO^- \tag{8.24}$$

$$HClO^- + H^+ \rightleftharpoons Cl\cdot + H_2O \tag{8.25}$$

$$Cl^- + Cl\cdot \rightleftharpoons Cl_2^- \tag{8.26}$$

These equilibria are pH-dependent, and the reactions may have important consequences for storage of nuclear waste in salt media.

The oxidation of Fe^{2+} to Fe^{3+} in acidic oxygenated solution is the basis for the *Fricke dosimeter*, §8.10. The reaction scheme is:

$$e_{aq}^- + H^+ \rightarrow H\cdot \tag{8.27}$$